

Validation Depth Calibration Module (VDCM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VALIDOS® — Validation Governance Architecture

Parent Standard: Validation Purpose & Scope Standard (VPSS)

Operational Layer: Validation Purpose & Scope Governance Layer

Category: AI & Interpretation

Subcategory: Validation Governance

Type: Parent Standard Module

Governed Space: Validation Depth Calibration

Version: 1.0

Status: Canonical · Open Module

Effective Date: 7 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS®
· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Validation Depth Calibration

Canonical Definition

Validation Depth Calibration Module (VDCM) defines the governance framework for determining, calibrating, justifying, documenting, and maintaining the degree of validation examination required to satisfy an approved Validation Purpose within a defined Validation Scope.

It establishes the conditions under which validation depth is proportionately aligned with intended use, operational significance, complexity, uncertainty, potential consequences, dependency exposure, reversibility, governance requirements, and risk.

Operational Role

VDCM governs the depth-calibration layer of Validation Purpose & Scope governance.

Its role is to determine **how extensively and rigorously the Validation Object must be examined** within the approved Validation Scope to provide sufficient validation for the stated Validation Purpose.

VDCM prevents validation depth from being determined arbitrarily,

uniformly, or solely by technical convenience.

Module Operational Space

VDCM governs:

- validation depth,
 - depth calibration,
 - validation intensity,
 - validation proportionality,
 - examination granularity,
 - validation rigor,
 - consequence sensitivity,
 - uncertainty sensitivity,
 - complexity sensitivity,
 - dependency sensitivity,
 - risk-informed depth,
 - depth justification,
 - depth escalation,
 - depth reduction,
 - depth reassessment,
 - and depth traceability.
-

Module Function

VDCM applies after the Validation Purpose and Validation Scope have been sufficiently established.

Its function is to prevent:

- validation that is too shallow for its intended purpose,
 - unnecessary validation intensity without methodological justification,
 - identical validation depth being applied to materially different risk environments,
 - high-consequence reliance based on low-depth examination,
 - validation depth being determined solely by available resources,
 - hidden reductions in validation rigor,
 - and validation conclusions implying a greater degree of examination than actually occurred.
-

Validation Depth Principle

Validation depth should be proportionate to the governance need.

More validation is not automatically better validation.

Less validation is not automatically efficient validation.

The appropriate depth is the level of examination necessary to produce a validation determination capable of supporting the approved purpose under the defined conditions.

VDCM therefore governs:

Depth Determination Factors

Validation depth may be influenced by multiple factors.

Intended Use

Validation supporting consequential operational deployment may require greater depth than validation supporting exploratory internal analysis.

Operational Significance

Objects performing critical or systemically important functions may require more extensive examination.

Potential Consequences

The severity and scale of potential harm, failure, error, or misreliance may increase required validation depth.

Complexity

Highly interconnected, adaptive, opaque, or multi-component Validation Objects may require deeper examination than simple and stable objects.

Uncertainty

Greater uncertainty regarding behavior, evidence, operating conditions, dependencies, or expected outcomes may require increased validation depth.

Dependency Exposure

Objects materially dependent on external systems, datasets, infrastructure, actors, or environments may require deeper examination of relevant dependency conditions.

Reversibility

Where consequences are difficult or impossible to reverse, validation may require greater depth before reliance is permitted.

Governance and Regulatory Requirements

Applicable governance, legal, regulatory, contractual, institutional, or assurance requirements may establish minimum validation-depth expectations.

Risk

Risk information may inform depth calibration without replacing

Depth Calibration

VDCM does not require a universal validation-depth scale for every domain.

Validation depth may be expressed through:

- defined validation levels,
- examination intensity,
- required evidence thresholds,
- required methodological diversity,
- sampling requirements,
- testing coverage,
- scenario coverage,
- review intensity,
- replication requirements,
- or other domain-appropriate depth mechanisms.

The selected mechanism must remain explicit, governable, and traceable to the approved Validation Purpose and Scope.

Minimum Implementation Framework

1. Establish Depth Drivers

Identify the factors materially affecting required validation depth, including where applicable:

- intended use,
- significance,
- consequences,
- complexity,
- uncertainty,
- dependencies,
- reversibility,
- governance requirements,
- and risk.

2. Determine Required Depth

Establish the level of examination necessary to satisfy the Validation Purpose across the approved Validation Scope.

3. Justify Depth

Document why the selected validation depth is proportionate to the governance need.

The justification should be capable of explaining why a materially

lower depth would be insufficient and why materially greater depth is or is not required.

4. Align Subsequent Validation Requirements

Ensure that subsequent validation criteria, methods, evidence requirements, execution intensity, and determination logic can support the approved validation depth.

5. Preserve Depth Traceability

Maintain:

- depth determination,
- depth drivers,
- depth justification,
- assumptions,
- depth changes,
- escalation or reduction decisions,
- relevant risk or governance references,
- timestamps,
- and sufficient traceability throughout the validation lifecycle.

Depth Escalation

Validation depth may need to increase where new information reveals:

- greater operational significance,
- higher uncertainty,
- additional dependencies,
- increased potential consequences,
- previously unknown complexity,
- broader intended reliance,
- new regulatory requirements,
- or materially changed risk.

Depth escalation must propagate to affected subsequent validation activities.

Depth Reduction

Validation depth may be reduced where justified by governed evidence or changed conditions.

A reduction should not occur solely because:

- validation is expensive,
- validation takes too long,
- required information is difficult to obtain,
- available resources are limited,
- or operational pressure favors faster completion.

Resource constraints may affect implementation decisions, but they do not independently establish methodological sufficiency.

Where required depth cannot be achieved, the validation purpose, scope, applicability, determination, or permitted reliance may need to be constrained.

Depth Reassessment

Validation depth should be reassessed when material changes occur in:

- Validation Purpose,
- Validation Scope,
- intended use,
- operating environment,
- risk,
- object complexity,
- uncertainty,
- dependencies,
- potential consequences,
- or intended reliance.

A previously sufficient depth may become insufficient when the conditions supporting the original determination change.

Validation Depth and Reliance

Validation depth and reliance are related but not identical.

VDCM determines the required degree of examination.

Later VALIDOS® governance determines whether and under which conditions reliance upon the resulting validation is justified.

However, the level of permitted reliance should never imply a validation depth greater than the depth actually performed.

Use Case 1 — AI Decision-Support System

Scenario

An AI system is initially validated for low-consequence internal analytical assistance and later proposed for use in decisions materially affecting customers.

Application

VDCM recognizes the change in intended use and potential consequences as depth drivers requiring reassessment.

The original validation depth may no longer be sufficient for the expanded purpose.

Result

Validation intensity is recalibrated before the previous validation can be treated as sufficient for the higher-consequence environment.

Use Case 2 — Critical Infrastructure Component

Scenario

A predictive system is used to support maintenance decisions for critical infrastructure.

Failure of the system could contribute to significant operational disruption.

Application

VDCM considers operational significance, consequence severity, dependency exposure, uncertainty, and reversibility when establishing the required validation depth.

Result

The validation depth is explicitly justified against the consequences of reliance rather than being determined by a generic validation procedure.

Canonical Closing Statement

Validation Depth Calibration Module establishes the canonical governance framework for determining how deeply a Validation Object must be examined within an approved Validation Scope. By governing proportionality, rigor, complexity, uncertainty, consequence sensitivity, dependency exposure, risk-informed calibration, escalation, reduction, reassessment, and traceability, VDCM ensures that validation depth remains proportionate to the purpose it must satisfy and the reliance it may ultimately be expected to support.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

→ [Previous module: Validation Scope Definition Module \(VSCDM\)](#)

→ [Next module: Validation Applicability Mapping Module \(VAMM\)](#)

Related Documents

→ [Validation Purpose & Scope Standard](#)

→ [About Validation Purpose & Scope Standard](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>